MTM VALUATION: FORWARDS VERSUS FUTURES

Learning Outcome

- compare the value and price of forward and futures contracts

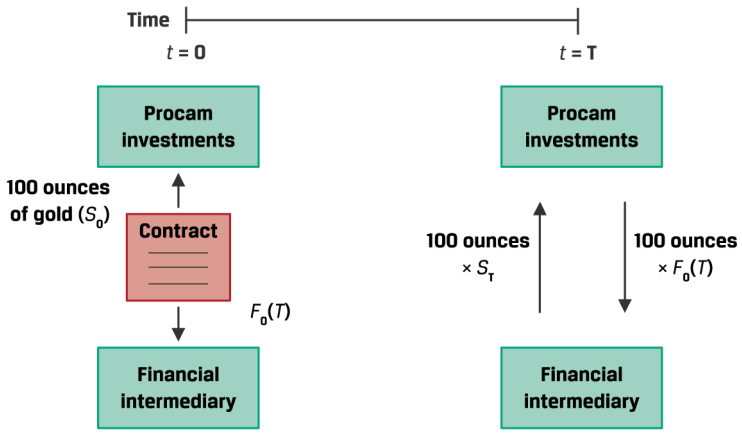
Examples 1 and 2 show the similarities between forward and futures prices at contract inception. Over time, different forward and futures contract features lead to different MTM values for contracts with the same underlying assets and otherwise identical details. Example 3 shows how the daily settlement of gains and losses causes this difference to arise.

EXAMPLE 3

We extend the earlier example to compare forward and futures pricing and valuation. In both cases, Procam Investments enters a cash-settled forward commitment to buy 100 ounces of gold at a price ($f_0[T] = F_0[T]$) of $1,778.76 per ounce in 91 days, with a risk-free rate of 2% and no gold storage cost.

Forward Contract

The forward price is $F_0(T)$ = $1,778.76 per ounce. No cash is exchanged or deposited at inception, and the contract value at inception, $V_0(T)$, is zero.



Over time, the forward price, $F_0(T)$, does not change, and the MTM at any time, [$V_t(T)$], equals the difference between the current spot price, S_t, and the present value of the forward price, PV_t of $F_0(T)$, shown from Procam's (the forward buyer's) perspective:

$$V_t(T) = S_t - F_0(T)(1 + r)^{-(T-t)}.$$

For example, say 71 days have elapsed and 20 days remain to maturity, $T - t$ = 20/365 or 0.0548. If the gold spot price, (S_t), has fallen by $50 since inception to $1,720 per ounce, solve for $V_t(T)$ as follows:

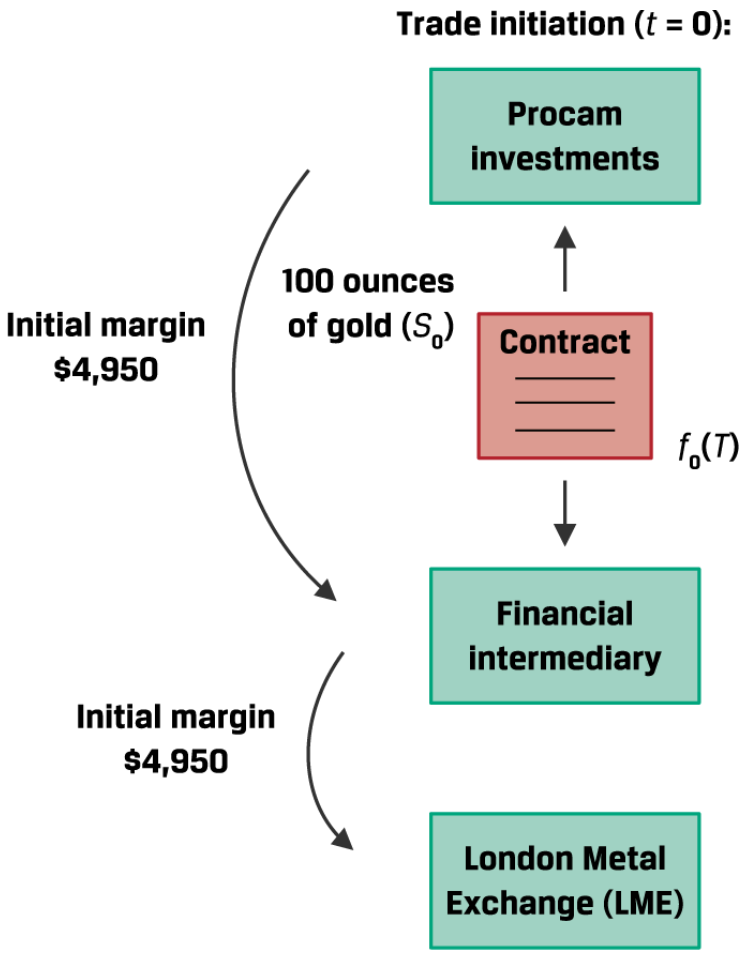
$$V_t(T) = \$1{,}720 - \$1{,}778.76(1.02)^{-0.0548}$$

= –$56.83 per ounce, or a $5,683 MTM loss (= –$56.83 × 100 ounces).

Under terms of the forward contract, no settlement of the MTM amount occurs until maturity. This process of resetting the contract value to zero each day makes it very unlikely that the futures contract would reach a similar MTM value.

Futures Contract

$4,950 per gold contract (100 ounces) and maintain a maintenance margin of $4,500 per contract. If a margin balance falls below $4,500, a counterparty receives a margin call and must immediately replenish its account to the initial $4,950.



Consider the first day of trading, where the spot gold price, (S_0), is $1,770 per ounce and the opening gold futures price, $f_0(T)$, is $1,778.76 per ounce.

- Assume that the gold futures price, $f_1(T)$, falls by $5 on the first trading day to $1,773.76 and the spot price, S_1, ends the day at a no arbitrage equivalent of $1,765.12 (= $1,773.76(1.02)$^{-90/365}$).
- Procam realizes a $500 MTM loss (= $5 per ounce × 100 ounces) deducted from its margin account, leaving Procam with $4,450.
- The MTM value of Procam's futures contract resets to zero at the futures closing price of $1,773.76 per ounce.
- Since Procam's margin account balance has fallen below the $4,500 maintenance level, it must deposit $500 to return the balance to the $4,950 initial margin.

Futures versus Forward Price and Value over Time

Using the same details, we compare the futures and forward price and value over two trading days. Assume that day two trading opens at day one's closing spot and futures prices. The following table shows the comparison:

Beginning of Day 2 Trading

Contract Type	Contract Price	Contract MTM	Realized MTM	Margin Deposit
Forward	$F_0(T)$ = $177,876	–$498	$0	$0
Futures	$f_1(T)$ = $177,376	$0	–$500	$4,950

$PV_t[F_0(T)]$, here with 90 days remaining to maturity, $T - t = 90/365$ or 0.24657:

$$V_t(T) = \$1{,}765.12 - \$1{,}778.76(1.02)^{-0.24657}$$

$$= \$4.98 \text{ per ounce.}$$

Extending our example to the beginning of day three at the prior day's close, assume a \$4 per ounce fall in the gold futures price on day two, $f_2(T)$, to \$1,769.76 and a no arbitrage equivalent spot price decline, S_2, to \$1,761.24 (= $\$1{,}769.76[1.02]^{-89/365}$). The following table shows the summary on day three:

Beginning of Day 3 Trading

Contract Type	Contract Price	Contract MTM	Realized MTM	Margin Deposit
Forward	$F_0(T)$ = \$177,876	–\$895	\$0	\$0
Futures	$f_2(T)$ = \$176,976	\$0	–\$400	\$4,550

The forward MTM contract value, $V_t(T)$, equals the difference between the current spot price, S_2 = \$1,761.24, and the present value of the original forward price, $PV_t[F_0(T)]$, here with 89 days remaining to maturity, $T - t = 89/365$ or 0.24384:

$$Vt(T) = \$1{,}761.24 - \$1{,}778.76(1.02)^{-0.24384}$$

$$= \$8.95 \text{ per ounce.}$$

Example 3 demonstrates the key differences in the price and value of forward and futures contracts over time. The forward contract price, $F_0(T)$, remains fixed until the contract matures. Forward contract MTM value changes are captured by the difference between the current spot price, S_t, and the present value of the forward price, $PV_t[F_0(T)]$. This forward contract MTM is not settled until maturity, giving rise to counterparty credit risk over time since no cash is exchanged from inception of the contract to its maturity or expiration. Futures contract prices fluctuate daily based upon market changes. The daily settlement mechanism resets the futures MTM to zero, and variation margin is exchanged to settle the difference, reducing counterparty credit risk. The *cumulative* realized MTM gain or loss on a futures contract is approximately the same as for a comparable forward contract. We will explore these differences in the next lesson after first turning our attention to forward and futures contracts on market reference rates.

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